1.2.12

EE25BTECH11012-BEERAM MADHURI

Question:
For $|x| < 1$, $\coth(x)$ can be approximated as: (EC 2007)

a) x
b) x^2
c) $\frac{1}{x}$
d) $\frac{1}{x^2}$

Solution:
The hyperbolic cotangent function is defined as:

$$\coth(x) = \frac{\cosh(x)}{\sinh(x)} \tag{0.1}$$

To find the approximation for small x, we use the Taylor Series expansions for $\cosh(x)$ and $\sinh(x)$ centered at 0: Expansion of cosh(x) and sinh(x)

$$\cosh(x) : 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \tag{0.2}$$

$$\sinh(x) : x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \tag{0.3}$$

Now, substitute these into the $\coth(x)$ formula:

$$\coth(x) = \frac{1 + \frac{x^2}{2} + \dots}{x + \frac{x^3}{6} + \dots} \tag{0.4}$$

For very small values of x, we can focus on the dominant terms (the lowest power of x in the numerator and denominator):

$$\coth(x) \approx \frac{1}{x} \tag{0.5}$$

Correct Option: c) $\frac{1}{x}$

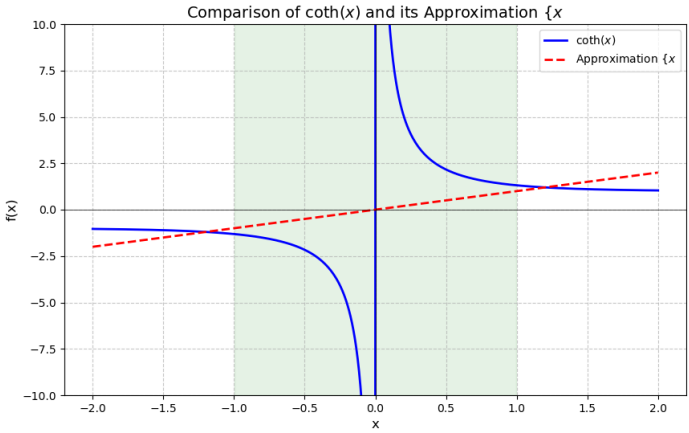

Fig. 0.1: Coth(x) and x comparision for $|x| < 1$

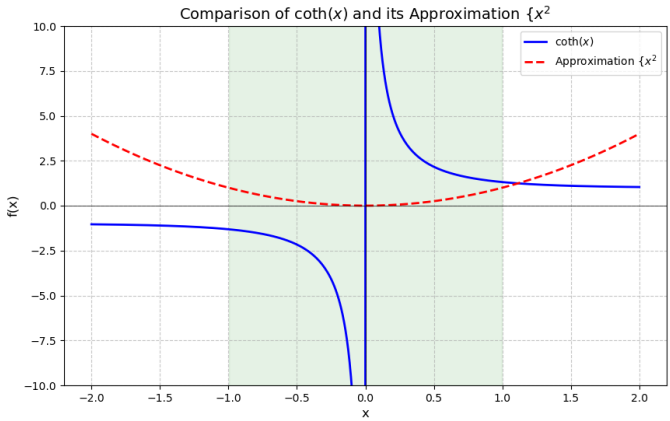

Fig. 0.2: Coth(x) and x^2 comparision for $|x| < 1$

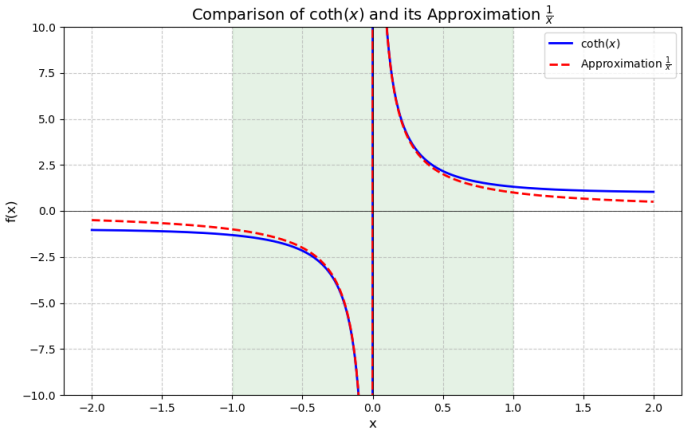

Fig. 0.3: Coth(x) and 1/x comparision for $|x| < 1$

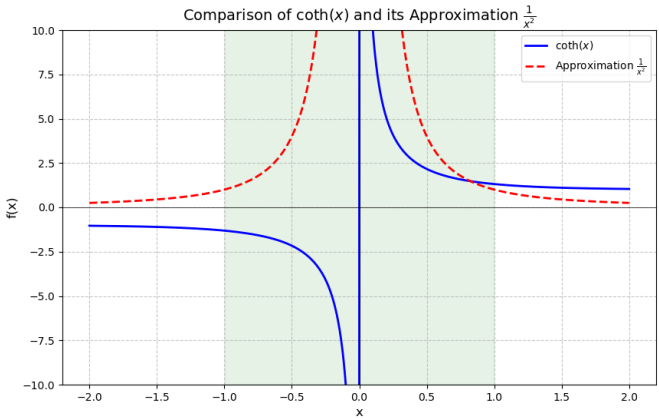

Fig. 0.4: Coth(x) and $1/x^2$ comparision for $|x| < 1$